You scratch my back, I'll ride on yours

p. 225 … it seems to be only in the social insects that [the evolution of sterile workers] has actually happened.

That is what we all thought. We had reckoned without naked mole rats. Naked mole rats are a species of hairless, nearly blind little rodents that live in large underground colonies in dry areas of Kenya, Somalia, and Ethiopia. They appear to be truly 'social insects' of the mammal world. Jennifer Jarvis's pioneering studies of captive colonies at the University of Capetown have now been extended by the field observations of Robert Brett in Kenya; further studies of captive colonies are being made in America by Richard Alexander and Paul Sherman. These four workers have promised a joint book, and I, for one, eagerly await it. Meanwhile, this account is based upon reading what few papers have been published and listening to research lectures by Paul Sherman and Robert Brett. I was also privileged to be shown the London Zoo's colony of naked mole rats by the then Curator of Mammals, Brian Bertram.

Naked mole rats live in extensive networks of underground burrows. Colonies typically number 70 or 80 individuals, but they can increase into the hundreds. The network of burrows occupied by one colony can be two or three miles in total length, and one colony may excavate three or four tons of soil annually. Tunnelling is a communal activity. A face worker digs at the front with its teeth, passing the soil back through a living conveyor belt, a seething, scuffling line of half a dozen little pink animals. From time to time the face-worker is relieved by one of the workers behind.

Only one female in the colony breeds, over a period of several years. Jarvis, in my opinion legitimately, adopts social insect terminology and calls her the queen. The queen is mated by two or three males only. All the other individuals of both sexes are nonbreeding, like insect workers. And, as in many social insect species, if the queen is removed some previously sterile females start to come into breeding condition and then fight each other for the position of queen.

The sterile individuals are called 'workers', and again this is fair enough. Workers are of both sexes, as in termites (but not ants, bees, and wasps, among which they are females only). What mole rat workers actually do depends on their size. The smallest ones, whom Jarvis calls 'frequent workers', dig and transport soil, feed the young, and presumably free the queen to concentrate on childbearing. She has larger litters than is normal for rodents of her size, again reminiscent of social insect queens. The largest nonbreeders seem to do little except sleep and eat, while intermediate-sized nonbreeders behave in an intermediate manner: there is a continuum as in bees, rather than discrete castes as in many ants.

Jarvis originally called the largest nonbreeders nonworkers. But could they really be doing nothing? There is now some suggestion, both from laboratory and field observations, that they are soldiers, defending the colony if it is threatened; snakes are the main predators. There is also a possibility that they act as 'food vats' like 'honey-pot' ants (see p. 223). Mole rats are homocoprophagous, which is a polite way of saying that they eat one another's faeces (not exclusively: that would run foul of the laws of the universe). Perhaps the large individuals perform a valuable role by storing up their faeces in the body when food is plentiful, so that they can act as an emergency larder when food is scarce—a sort of constipated commissariat.

To me, the most puzzling feature of naked mole rats is that, although they are like social insects in so many ways, they seem to have no equivalent caste to the young winged reproductives of ants and termites. They have reproductive individuals, of course, but these don't start their careers by taking wing and dispersing their genes to new lands. As far as anyone knows, naked mole rat colonies just grow at the margins by expanding the subterranean burrow system. Apparently they don't throw off long-distance dispersing individuals, the equivalent of winged reproductives. This is so surprising to my Darwinian intuition that it is tempting to